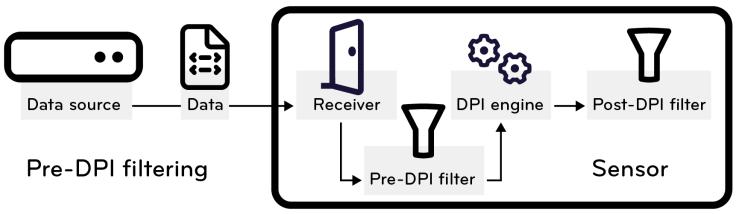
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Pre-DPI Filters 5.5.0

You can optimize sensor performance by filtering unnecessary traffic before doing Deep Packet Inspection (DPI). To configure this, you create a filter to accept or drop traffic based on the IP addresses, protocols, ports, and other packet attributes that are present in traffic. Unlike Traffic Filters, which apply rules after the DPI engine inspects all packets, pre-DPI filters streamline processing by discarding unnecessary data earlier. This reduces overhead and increases throughput.



The pre-DPI filter rules that Stellar Cyber uses are based on expressions that packet capture tools like tcpdump and Wireshark employ to define which network traffic to capture or exclude. Stellar Cyber follows the same syntax to determine which traffic to include or exclude before performing DPI, thereby conserving processing resources. For more information about filtering expressions and their syntax, see TCPDUMP & LIBPCAP.

Term	Definition	Usage in Stellar Cyber	Example
Expression	A statement in valid tcpdump syntax that defines one or more conditions for filtering traffic.	Multiple conditions in an expression are combined by using the operators `and`, `or`, and `not`.	`src host 10.0.0.1 and src port 443` (one expression) `dst host 192.168.1.1` (another expression)
Rule	A set of one or more expressions with a directive to **Accept** or **Drop** traffic.	Multiple rules in a filter are joined by using the operator `or`.*	`(src host 10.0.0.1 and src port 443) or (dst host 192.168.1.1)`

* In Stellar Cyber, you can create a filter by entering one or more rules, each on its own line, and letting Stellar Cyber implicitly add `or` between them, or you can define a rule by placing it inside parentheses and explicitly entering `or` between each parenthetically enclosed rule in a single line.

To access the Pre-DPI Filter option, select **System | DATA MANAGEMENT | Sensors | Sensor Profiles | Create | Add Modular Sensor Profile** or **Add Standard Sensor Profile** and then enable **Network Traffic** and **Traffic Filtering**. You can also access it by editing a previously created sensor profile.

Common Pre-DPI Filter Rules

Depending on the action you choose—**Accept** or **Drop**—the following rules define traffic attributes that must be present for a Stellar Cyber sensor to either accept or drop traffic before performing Deep Packet Inspection (DPI).

Filter by IP Address

This rule matches traffic originating from 10.36.22.54:

```
src host 10.36.22.54
```

This rule matches traffic destined for 192.168.1.1:

```
dst host 192.168.1.1
```

This rule matches traffic from or to 192.168.1.112:

```
host 192.168.1.112
```

The behavior of a rule in a pre-DPI filter is different if you specify the source or destination for `host`, `ip`, or `port` or if you don't specify it. The different behaviors depend on whether the traffic is sensor-initiated or pass-through:

- **Sensor-initiated traffic**: `src { host | ip | port }` and `dst { host | ip | port }` do not take effect. Use `host`, `ip`, or `port`, which applies regardless of direction.
- **Pass-through traffic**: `src { host | ip | port }` and `dst { host | ip | port }` work correctly, filtering based on traffic source or destination. `host`, `ip`, and `port` also work and are applied whether they are the source or destination.

Because the pre-DPI filter is stateless, you must define rules for both directions when you want to accept or drop bidirectional pass-through traffic. For example, to block DNS traffic between a subnet (10.1.1.0/24) and a DNS server (192.168.1.12), define a filter with rules for both directions as shown below and set the action to **Drop**. The filter automatically applies an `or` operator between each rule:

```
src net 10.1.1.0/24 and dst host 192.168.1.12 and port 53
```

```
src host 192.168.1.12 and dst net 10.1.1.0/24 and port 53
```

Filter by Network Range (CIDR)

This rule matches traffic for the entire 192.168.1.0/24 subnet:

```
net 192.168.1.0/24
```

Filter by Protocol

This rule matches only TCP packets:

```
tcp
```

This rule matches only UDP packets:

```
udp
```

Filter by Port

This rule matches traffic on port 443 (HTTPS):

```
port 443
```

This rule matches traffic sent to port 22 (SSH):

```
dst port 22
```

Combine Conditions and Rules with Logical Operators

When using the Pre-DPI filter in Stellar Cyber, you can enter one or more filtering rules in a single line and in multiple lines:

- **Single line**, where you enter a valid expression to express one or more rules and explicitly join them by entering the logical operator `or` between parenthetically enclosed rules.
- **Multiple lines**, where each new line represents one or more rules and each line is implicitly joined by `or` within the filter.

Understanding how these operators work is essential for creating effective filters:

- When you combine conditions using `and`, all specified conditions must be met for the rule to apply.
- When you combine conditions using `or`, the rule applies if any of the conditions are met.
- When you combine conditions using `not`, the rule excludes traffic that matches the specified condition.

  Use `not` at the beginning of a rule to negate the entire condition. Use `and not` within an expression to match the first condition but exclude the second.

When entering one or more rules in a single line, you can combine multiple conditions with `and`, `or`, or `not` in a parenthetically enclosed rule and then combine these rules with `or`.

When entering rules on separate lines, you can combine multiple conditions with `and`, `or`, or `not` on each line. The filter then automatically joins all rules with `or`.

With either approach—rules enclosed in parentheses and explicitly joined by `or` or rules entered individually on their own lines and implicitly joined by `or`--traffic matches the filter if it meets the conditions defined in any of its rules.

Filters do not work if the operators are uppercase `AND`, `OR`, `NOT`. They must be lowercase `and`, `or`, `not` to take effect.

Example: Create a Filter with Two Rules

In this example, you want to drop HTTP and HTTPS traffic (ports 80 and 443) from 10.0.0.1 and all traffic from 10.0.0.2. After setting the action as **Drop**, create a filter consisting of either two rules in the same line or two individual rules on separate lines.

Option 1: Use a single line

If you prefer entering both rules on a single line, put each rule in parentheses and explicitly set `or` as the logical operator between them:

```
(src host 10.0.0.1 and (src port 80 or src port 443)) or (src host 10.0.0.2)
```

These rules state that traffic from 10.0.0.1 is only dropped if it is on port 80 or 443 while all traffic from 10.0.0.2 is dropped regardless of port.

Option 2: Use multiple lines

Instead of entering multiple rules on a single line, you can enter each rule on its own line:

```
src host 10.0.0.1 and (src port 80 or src port 443)
```

```
src host 10.0.0.2
```

The rules in the second option achieve the same result as those in the first option, while making the filter a little easier to read.

Example: Create a Filter with Three Rules

In this example, you want to accept TCP traffic from the 192.168.1.0/24 network, all traffic except SSH (port 22) from 10.0.0.5, and all traffic destined for 172.16.0.1 regardless of source or protocol. After setting the action as **Accept**, create a filter consisting of either three rules on a single line or three individual rules on separate lines.

Option 1: Use a single line

If you prefer entering all three rules on the same line, put each rule in parentheses and explicitly set `or` as the logical operator between them:

```
(net 192.168.1.0/24 and tcp) or (src host 10.0.0.5 and not port 22) or (dst host 172.16.0.1)
```

By enclosing each rule in parentheses and joining the rules with `or`, the filter ensures that traffic is accepted if it matches any of the rules.

Option 2: Use multiple lines

Instead of entering all three rules on a single line, you can enter each rule on its own line:

```
net 192.168.1.0/24 and tcp
```

```
src host 10.0.0.5 and not port 22
```

```
dst host 172.16.0.1
```

The rules in the second option achieve the same result as those in the first, while making the individual rules that constitute the filter a little clearer to read.

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